

Linear Transformations and Their Matrices

Lecture Notes

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Introduction

This lecture introduces the fundamental concept of a **linear transformation**. While much of linear algebra focuses on matrices, a linear transformation is a more abstract, "coordinate-free" idea that describes how a vector space is mapped to another. We will explore what makes a transformation linear, see intuitive examples, and then establish the crucial link: every linear transformation can be represented by a matrix once we choose a basis.

The core goal is to understand that matrices are the computational tools that represent these more fundamental geometric and algebraic operations.

1 What is a Linear Transformation?

A **transformation** (or mapping) T is a function that takes an input vector $\mathbf{v}$ from one vector space (the *domain*) and produces an output vector $T(\mathbf{v})$ in another vector space (the *codomain*).

While there are many kinds of transformations, in linear algebra, we are interested in those that preserve the vector space operations of addition and scalar multiplication.

1.1 Definition of a Linear Transformation

💡 Key Idea

A transformation T is **linear** if it satisfies two rules for all vectors $\mathbf{v}$ and $\mathbf{w}$ in its domain and for any scalar c :

1. **Additivity:** $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$
(The transformation of a sum is the sum of the transformations.)
2. **Homogeneity:** $T(c\mathbf{v}) = cT(\mathbf{v})$
(The transformation of a scaled vector is the scaled transformation of the vector.)

These two rules can be combined into a single defining property:

$$T(c\mathbf{v} + d\mathbf{w}) = cT(\mathbf{v}) + dT(\mathbf{w}) \quad \text{for any scalars } c \text{ and } d.$$

A crucial consequence of this definition is that for any linear transformation, the zero vector must map to the zero vector: $T(\mathbf{0}) = \mathbf{0}$.

2 Examples and Non-Examples

2.1 Linear Examples (Coordinate-Free)

1. **Projection:** Consider a line passing through the origin in $\mathbb{R}^2$. The projection T maps every vector $\mathbf{v}$ onto its closest point on that line. This is linear because if you double $\mathbf{v}$, its projection doubles, and the projection of a sum is the sum of projections.
2. **Rotation:** The transformation T rotates every vector $\mathbf{v}$ in the plane by a fixed angle (e.g., 45°) around the origin. This is linear as rotating a sum of vectors is the same as adding the rotated vectors.

2.2 Non-Linear Examples

1. **Shifting the Plane (Translation):** The transformation is $T(\mathbf{v}) = \mathbf{v} + \mathbf{v}_0$ for a fixed vector $\mathbf{v}_0 \neq \mathbf{0}$. This is not linear because $T(\mathbf{0}) = \mathbf{v}_0$, which violates the necessary condition $T(\mathbf{0}) = \mathbf{0}$.
2. **Taking the Length of a Vector:** The transformation is $T(\mathbf{v}) = \|\mathbf{v}\|$. This is not linear because it fails homogeneity for negative scalars. For example, $T(-\mathbf{v}) = \|-\mathbf{v}\| = \|\mathbf{v}\|$, but $-1 \cdot T(\mathbf{v}) = -\|\mathbf{v}\|$. Since $\|\mathbf{v}\| \neq -\|\mathbf{v}\|$ (for $\mathbf{v} \neq \mathbf{0}$), it is not linear.

3 The Bridge: Transformations from Matrices

💡 Key Idea

For any $m \times n$ matrix A , the transformation T defined by the rule $T(\mathbf{v}) = A\mathbf{v}$ is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.

This follows directly from the properties of matrix multiplication:

1. **Additivity:** $A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w}$
2. **Homogeneity:** $A(c\mathbf{v}) = c(A\mathbf{v})$

This shows that matrix multiplication is inherently a linear operation. For example, the matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ defines a linear transformation that reflects vectors in $\mathbb{R}^2$ across the x-axis.

4 The Core Concept: From a Transformation to a Matrix

The main goal is to reverse the process: given an abstract linear transformation T , how do we find the unique matrix A that represents it? The answer is: **we must choose a basis**.

4.1 The Role of a Basis

A linear transformation is completely determined by what it does to the basis vectors of the input space. If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis for the input space, any vector $\mathbf{v}$ can be written as a unique linear combination:

$$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n$$

The numbers $(c_1, \dots, c_n)$ are the **coordinates** of $\mathbf{v}$ in this basis. Because T is linear, we can find $T(\mathbf{v})$ easily:

$$T(\mathbf{v}) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + \dots + c_nT(\mathbf{v}_n)$$

This means if we know what T does to the basis vectors, we know everything.

4.2 The Algorithm to Construct Matrix A

To find the $m \times n$ matrix A representing $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$:

1. **Choose a basis for the input space $\mathbb{R}^n$:** $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$.
2. **Choose a basis for the output space $\mathbb{R}^m$:** $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$.
3. **Construct the columns of A one by one:**
 - **For the j -th column of A :**
 - a) Apply the transformation T to the j -th input basis vector, $\mathbf{v}_j$.
 - b) The result, $T(\mathbf{v}_j)$, is a vector in the output space. Express $T(\mathbf{v}_j)$ as a linear combination of the output basis vectors:

$$T(\mathbf{v}_j) = a_{1j}\mathbf{w}_1 + a_{2j}\mathbf{w}_2 + \dots + a_{mj}\mathbf{w}_m$$

- c) The coordinates $(a_{1j}, a_{2j}, \dots, a_{mj})$ form the **j -th column** of the matrix A .

This matrix A correctly transforms the *input coordinates* of a vector into its *output coordinates*.

5 Worked Examples of Constructing a Matrix

5.1 Example 1: Projection Transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$

Let T be the projection onto a line in $\mathbb{R}^2$.

Case A: Using a "Good" Basis (Eigenvectors)

Let's choose a basis that is natural for the problem.

- **Input/Output Basis:** $\{\mathbf{v}_1, \mathbf{v}_2\}$, where $\mathbf{v}_1$ is a unit vector **along** the line and $\mathbf{v}_2$ is a unit vector **perpendicular** to it.

1. **Find 1st Column:** $T(\mathbf{v}_1) = \mathbf{v}_1 = 1\mathbf{v}_1 + 0\mathbf{v}_2$. The coordinates are $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$.
2. **Find 2nd Column:** $T(\mathbf{v}_2) = \mathbf{0} = 0\mathbf{v}_1 + 0\mathbf{v}_2$. The coordinates are $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Resulting Matrix: In this ideal basis, the matrix for projection is diagonal: $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

Case B: Using the Standard Basis

Let's project onto the 45° line using the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$. The matrix becomes the standard projection matrix: $P = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. This matrix is not diagonal because the standard basis vectors are not eigenvectors of the projection.

5.2 Example 2: The Derivative as a Linear Transformation

Let T be the derivative operator d/dx .

- **Input Space:** Polynomials of degree ≤ 2 . Basis: $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \{1, x, x^2\}$.
- **Output Space:** Polynomials of degree ≤ 1 . Basis: $\{\mathbf{w}_1, \mathbf{w}_2\} = \{1, x\}$.

T maps a 3D space to a 2D space, so A will be 2×3 .

1. **1st Column:** $T(\mathbf{v}_1) = \frac{d}{dx}(1) = 0 = 0\mathbf{w}_1 + 0\mathbf{w}_2$. Coordinates: $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

2. **2nd Column:** $T(\mathbf{v}_2) = \frac{d}{dx}(x) = 1 = 1\mathbf{w}_1 + 0\mathbf{w}_2$. Coordinates: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

3. **3rd Column:** $T(\mathbf{v}_3) = \frac{d}{dx}(x^2) = 2x = 0\mathbf{w}_1 + 2\mathbf{w}_2$. Coordinates: $\begin{bmatrix} 0 \\ 2 \end{bmatrix}$.

Resulting Matrix: $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$. If we apply this to the input coordinate vector $\begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$,

we get $\begin{bmatrix} c_2 \\ 2c_3 \end{bmatrix}$, which are the coordinates of the correct derivative $c_2 + 2c_3x$.

6 Summary & Applications

- **Key Idea:** Linear transformations are the fundamental, coordinate-free operations on vector spaces. Matrices are their concrete representations once bases are chosen.
- **The Process:** To find the matrix A for a transformation T , you must know how T acts on the input basis vectors. The coordinates of these transformed vectors (in the output basis) form the columns of A .
- **Significance:** This connection is why matrix algebra is so powerful. Matrix multiplication corresponds to the composition of linear transformations, and a matrix inverse corresponds to the inverse transformation.

Practical Applications in Data Science & ML

- **Principal Component Analysis (PCA):** PCA is a linear transformation that re-expresses data in a new basis (the principal components). This new basis is the "good" basis, like the eigenvector basis for projection, where the resulting transformation matrix is simple and reveals the most important features of the data.
- **Neural Networks:** Each layer in a standard feed-forward neural network performs an affine transformation, which is a linear transformation ($y = Wx$) followed by a shift ($+b$). The "learning" process is essentially finding the optimal matrices W (weights) for each layer.
- **Computer Graphics and Robotics:** The movement and orientation of objects are controlled by sequences of linear transformations (rotations, scaling, shearing). Finding the single matrix for a composite transformation is a common and essential task.

7 Exercise Problems

1. A linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ rotates vectors counter-clockwise by 90° . Using the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ for both input and output, find the matrix A that represents T .
2. Consider the linear transformation T from Example 1 (projection onto the 45° line). If you use the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ for the input but the "good" basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ (where $\mathbf{v}_1$ is on the 45° line and $\mathbf{v}_2$ is perpendicular) for the output, what is the resulting matrix A ?
3. Let $T : P_2 \rightarrow \mathbb{R}^2$ be a linear transformation that evaluates a polynomial $p(x) = a + bx + cx^2$ at $x = 0$ and $x = 1$. So, $T(p(x)) = \begin{bmatrix} p(0) \\ p(1) \end{bmatrix}$. Using the basis $\{1, x, x^2\}$ for P_2 and the standard basis for $\mathbb{R}^2$, find the matrix for T .
4. Is the transformation $T(\mathbf{v}) = \mathbf{v} + \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ a linear transformation on $\mathbb{R}^2$? Justify your answer in one sentence.